

UEI601: INDUSTRIAL INSTRUMENTATION

L	T	P	Cr
3	1	2	4.5

Course objectives: To provide the knowledge for the measurement of length, angle and area. To familiarize with motion and vibration measurement. To explain different methods for pressure and flow measurement. To introduce different methods of temperature, level and humidity measurement.

Pressure Measurement: Moderate pressure measurement, Bourdon tube, Bellows and diaphragms, High pressure measurement: Piezoelectric, Electric resistance, Low pressure measurement: Mcleod gauge, Pirani Gauge, Thermal conductivity, Ionization gauge, Dead weight gauges, MEMS-based pressure sensors, Wireless pressure sensors

Flow Measurement: Obstruction meter, Orifice, Nozzle, Venturi, Pitot tube, Rotameter, Turbine, Electromagnetic, Vortex, Positive displacement, Coriolis mass flow meters, Anemometers, Weirs and flumes, Laser Doppler anemometer, Ultrasonic flow meter, Mass flow meter.

Temperature Measurement: Resistance Temperature Detector, Thermocouples, Cold junction compensation, Bimetallic thermometers, Liquid-in-glass, Pressure thermometer, Semiconductor sensors, Digital thermometers, Pyrometers.

Level Measurement: Visual level indicators, Purge method, Buoyancy method, Resistance, Capacitance and inductive probes, Ultrasonic, Laser, Optical fiber, Thermal, Radar, Radiation.

Miscellaneous Measurements: Accelerometers, Humidity, Viscosity, Dimensional measurement, Gauge blocks, Density measurement, laser and sonar distance measurement, Tachometer for speed measurement. Advanced Sensors and Transducers, Smart Sensors, Biosensors, Wearable Sensors

Laboratory work: Experiments around Measurement of Pressure, Flow, Temperature, Level, Length, Humidity, Vibration, Viscosity using different techniques.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Analyze the construction & working of various industrial devices used to measure pressure and flow
2. Distinguish the different methods for level and viscosity measurement.
3. Elucidate different methods for temperature measurement.
4. Explicate the construction and working of advanced sensors and transducers

Text Books:

- 1 Doeblin, E.O., *Measurement systems, Applications and Design*, McGraw Hill (2004).
- 2 Nakra, B. C. and Chaudhry, K. K., *Instrumentation Measurement and Analysis*, McGraw Hill (2009)

Reference Books:

- 1 Murthy, D.V.S., *Transducers and Instrumentation*, Prentice Hall of India Private Limited (2003).

- 2 Sawhney, A.K., *A Course in Electrical and Electronic Measurements and Instrumentation*, Dhanpat Rai and Co. (P) Ltd. (2007).

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes/Lab Evaluations/Project)	30